

Reg. No.														
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 RAJALAKSHMI ENGINEERING COLLEGE An AUTONOMOUS Institution Affiliated to ANNA UNIVERSITY, Chennai	Continuous Assessment Test-I [CAT - I]
	Year : II
	Semester : III
	Branch : B.E/B.Tech Programmes
	Sub. Code : CS23334
	Subject Name : Fundamentals of Data Science
QP Code :	

[Regulations 2023]
Date: _____ **Time: 120 Minutes** **Marks: 75**
Answer ALL Questions
Part A [4 x 2 = 8 Marks]

1.1.	How can the use of unstructured data improve customer sentiment analysis?	[A2]	CO1
1.2.	How would a Data Scientist apply machine learning algorithms to predict customer churn in a telecom company?	[A2]	CO1
1.3.	How can data science be applied to enhance predictive maintenance in manufacturing?	[A2]	CO1
1.4.	In what way could semi-structured data be utilized to enhance search engine algorithms?	[A2]	CO1

Part B [2 x 11 = 22 Marks]

1.5.	a.	Analyze the importance of multi-factor authentication (MFA) in protecting user accounts. Discuss the various factors used in MFA, implementation strategies, and the specific threats it mitigates.	[B2]	CO1
		[OR]		
	b.	Analyze the responsibilities of various data science roles intersect, and what implications does this have for collaborative projects?	[B2]	CO1
1.6.	a.	Analyze the role of metadata in enhancing the usability and integrity of various types of data and also discuss how do different types of data impact the choice of data analysis techniques and tools?	[B2]	CO1
		[OR]		
	b.	Analyze the role of a Data Scientist in developing a customer segmentation model for a retail company. Discuss the steps involved, the techniques used, and how the results can be interpreted and applied to marketing strategies.	[B2]	CO1

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

2.1.	Write a Python script to discretize the Rating column of a DataFrame into three categories: 'Low', 'Medium', and 'High'.	[A2]	CO2
2.2.	How do you determine the number of bins to use when discretizing a continuous variable?	[A2]	CO2

2.3.	Calculate the mean, median, and mode of the Maths column. Students data.csv						[A2]	CO2
	Student_ID	Name	Age	Gender	Math	Science		
	1	Alice	14	F	85	78		
	2	Bob	15	M	90	88		
	3	Charlie	14	M	95	85		
	4	David	15	M	70	90		
	5	Eva	14	F	80	75		
	6	Frank	15	M	88	80		
	7	Grace	14	F	75	85		
	8	Helen	15	F	92	78		
	9	Ian	14	M	85	88		
	10	Jane	15	F	80	85		
2.4.	You have a dataset with missing values in multiple columns. Describe a strategy for imputing these missing values, considering both numerical and categorical data. What factors influence your choice of strategy?						[A2]	CO2

Part B [2 x 11 = 22 Marks]

2.5.	a.	In the following dataset of movies, what technique can be deployed to impute the release date feature? Provide the python code to carry out imputation.					[B2]	CO2		
		MovieID	MovieTitle	ReleaseDate	Genre	Rating			BoxOffice (Million \$)	Quarter of release year
		1	Movie A	2024-01-15	Action	7.8			150	Q1
		2	Movie B	2024-02-20	Comedy	6.5			80	Q1
		3	Movie C	NaN	Drama	8.2			200	Q1
		4	Movie D	2024-04-10	Action	7.5			120	Q2
		5	Movie E	NaN	Thriller	7.9			140	Q2
		6	Movie F	2024-03-25	Comedy	6.8			90	Q1
		7	Movie G	NaN	Action	7.4			160	Q2
		8	Movie H	2024-05-15	Drama	8.1			180	Q2
		9	Movie I	2024-06-01	Thriller	7.7			130	Q2
		10	Movie J	NaN	Comedy	6.9			85	Q2
[OR]										

	b. In the healthcare dataset given below, identify which fields need to be preprocessed for data security, data privacy and data integrity: <table><tr><th>Patient ID</th><th>Name</th><th>DOB</th><th>SSN</th><th>Address</th><th>Phone Number</th><th>Diagnosis</th><th>Date of Diagnosis</th><th>Treatment Cost (\$)</th><th>Last Visit Date</th><th>Emergency Contact Name</th><th>Emergency Contact Phone</th></tr><tr><td>1001</td><td>John Doe</td><td>1985-06-15</td><td>123-45-6789</td><td>123 Elm St, City</td><td>555-1234</td><td>Hypertension</td><td>2023-07-15</td><td>250.00</td><td>2024-08-15</td><td>Jane Doe</td><td>555-5678</td></tr><tr><td>1002</td><td>Alice Smith</td><td>1978-09-22</td><td>987-65-4321</td><td>456 Oak Ave, City</td><td>555-2345</td><td>Diabetes</td><td>2024-08-20</td><td>350.00</td><td>2024-08-20</td><td>Robert Smith</td><td>555-6789</td></tr><tr><td>1003</td><td>Bob Johnson</td><td>1990-03-10</td><td>111-22-3333</td><td>789 Pine Rd, City</td><td>555-3456</td><td>Asthma</td><td>2014-08-25</td><td>150.00</td><td>2024-08-22</td><td>Emily Johnson</td><td>555-7890</td></tr><tr><td>1004</td><td>Carol Brown</td><td>1982-12-05</td><td>444-55-6666</td><td>101 Maple Blvd, City</td><td>555-4567</td><td>Arthritis</td><td>2024-08-25</td><td>200.00</td><td>2024-08-25</td><td>Michael Brown</td><td>555-8901</td></tr><tr><td>1005</td><td>David Wilson</td><td>1965-11-30</td><td>777-88-9999</td><td>202 Birch St, City</td><td>555-5678</td><td>High Cholesterol</td><td>2023-08-27</td><td>300.00</td><td>2024-08-27</td><td>Sara Wilson</td><td>555-9012</td></tr></table>	Patient ID	Name	DOB	SSN	Address	Phone Number	Diagnosis	Date of Diagnosis	Treatment Cost (\$)	Last Visit Date	Emergency Contact Name	Emergency Contact Phone	1001	John Doe	1985-06-15	123-45-6789	123 Elm St, City	555-1234	Hypertension	2023-07-15	250.00	2024-08-15	Jane Doe	555-5678	1002	Alice Smith	1978-09-22	987-65-4321	456 Oak Ave, City	555-2345	Diabetes	2024-08-20	350.00	2024-08-20	Robert Smith	555-6789	1003	Bob Johnson	1990-03-10	111-22-3333	789 Pine Rd, City	555-3456	Asthma	2014-08-25	150.00	2024-08-22	Emily Johnson	555-7890	1004	Carol Brown	1982-12-05	444-55-6666	101 Maple Blvd, City	555-4567	Arthritis	2024-08-25	200.00	2024-08-25	Michael Brown	555-8901	1005	David Wilson	1965-11-30	777-88-9999	202 Birch St, City	555-5678	High Cholesterol	2023-08-27	300.00	2024-08-27	Sara Wilson	555-9012	[B2]	CO2
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2.6.	a. Use a flowchart to summarize the following procedures for attribute subset selection: (a) stepwise forward selection (b) stepwise backward elimination (c) a combination of forward selection and backward elimination	[B2]	CO2																																																																								
	[OR]																																																																										
	b. In the following dataset representing housing, which fields have scaling issues and which technique would you apply for those? <table><tr><th>ID</th><th>Crime Rate (per capita)</th><th>Property Tax Rate (\$10,000)</th><th>Avg Number of Rooms</th><th>Proportion of Units Built Prior to 1940</th><th>Median Value of Homes (\$1000s)</th></tr><tr><td>1</td><td>0.00001</td><td>10,000</td><td>6.575</td><td>65.2</td><td>24.0</td></tr><tr><td>2</td><td>0.00002</td><td>12,000</td><td>6.421</td><td>78.9</td><td>21.6</td></tr><tr><td>3</td><td>0.00003</td><td>15,000</td><td>7.185</td><td>61.1</td><td>34.7</td></tr><tr><td>4</td><td>0.00005</td><td>20,000</td><td>6.998</td><td>45.8</td><td>33.4</td></tr></table>	ID	Crime Rate (per capita)	Property Tax Rate (\$10,000)	Avg Number of Rooms	Proportion of Units Built Prior to 1940	Median Value of Homes (\$1000s)	1	0.00001	10,000	6.575	65.2	24.0	2	0.00002	12,000	6.421	78.9	21.6	3	0.00003	15,000	7.185	61.1	34.7	4	0.00005	20,000	6.998	45.8	33.4	[B2]	CO2																																										
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		5	0.00010	25,000	7.147	54.2	36.2		
		6	0.00007	18,000	6.430	65.0	28.7		
		7	0.00015	22,000	6.012	54.3	14.8		
		8	0.00020	30,000	6.072	47.9	20.6		
		9	0.00025	35,000	5.934	57.3	13.6		
		10	0.00030	40,000	6.379	62.8	22.6		

Answer ALL Questions

Part A [2 x 2 = 4 Marks]

3.1.	You are analyzing two datasets from different sources for the same research problem. Compare the significance of conducting Exploratory Data Analysis (EDA) on these datasets.	[A2]	CO3
3.2.	You decide to use Min-Max scaling and robust scaling on a dataset with extreme outliers. Analyze the impact of each scaling technique on the dataset's feature distribution.	[A2]	CO3

Part B [1 x 11 = 11 Marks]

3.3.	a.	You are given two datasets from different sources for the same research problem: one from a survey and one from a sensor network. Perform a comprehensive Exploratory Data Analysis (EDA) on both datasets. Your analysis should include the following: 1. Identify and discuss any data quality issues present in each dataset (e.g., missing values, outliers). 2. Compare and contrast the distribution of key variables in both datasets. 3. Discuss how the EDA results from each dataset might influence your approach to subsequent analysis or modeling. 4. Provide a summary of the insights gained from the EDA and any recommendations for handling the data moving forward.	[B2]	CO3
		[OR]		
	b.	You have been tasked with performing EDA on a complex dataset using a programming library (e.g., Matplotlib in Python). Your analysis should address:	[B2]	CO3

		1. Outline the steps you would take to perform EDA using tool. Include the types of visualizations you would create and the insights you aim to gain. 2. Give the effectiveness of each tool in terms of ease of use, depth of analysis, and ability to uncover patterns in the data. 3. Discuss any limitations you encountered with tool and how you addressed these limitations during your EDA. 4. Based on your experience with tool, provide an insight for different types of EDA tasks and explain why.		
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 RAJALAKSHMI ENGINEERING COLLEGE An AUTONOMOUS Institution Affiliated to ANNA UNIVERSITY, Chennai	Continuous Assessment Test- I [CAT - I]	
	Year	: II
	Semester	: III
	Branch	: B.E/B.Tech Programmes
	Sub. Code	: CS23334
Subject Name	: Fundamentals of Data Science	
QP Code	:	

[Regulations 2023]
Date: _____ **Time: 120 Minutes** **Marks: 75**
Answer ALL Questions
Part A [4 x 2 = 8 Marks]

1.1.	How can you apply qualitative data analysis to improve customer satisfaction in a retail store?	[A2]	CO1
1.2.	If the labels are known in a clustering project, how would you evaluate the performance of the model?	[A2]	CO1
1.3.	Describe a scenario where converting semi-structured data to structured data would be beneficial for an organization.	[A2]	CO1
1.4.	Explain how data science can be applied to improve student outcomes in education field.	[A2]	CO1

Part B [2 x 11 = 22 Marks]

1.5.	a.	Describe a detailed process for transforming unstructured customer feedback data into actionable insights to improve product development. Include steps for data collection, preprocessing, analysis, and application.	[B2]	CO1
		[OR]		
	b.	Analyze how a Machine Learning Engineer can deploy and monitor a machine learning model in production. Discuss the best practices for ensuring model reliability, performance, and maintenance.	[B2]	CO1
1.6.	a.	Analyze the role of regular security audits in maintaining data security within an organization. Discuss the steps involved in conducting a security audit, the types of audits, and the specific security issues they help identify and address.	[B2]	CO1
		[OR]		
	b.	Analyze the differences between the roles of a data scientist, data analyst, and data engineer in a large-scale data project.	[B2]	CO1

Answer ALL Questions

Part A [4 x 2 = 8 Marks]

2.1.	You are given a data set consisting of variables with more than 30 percent missing values. How will you deal with them?	[A2]	CO2
2.2.	Why is Python used for Data Cleaning in Data Science?	[A2]	CO2
2.3.	How would you design a data collection plan to gather user feedback on a new mobile app?	[A2]	CO2

2.4.	How would you integrate two datasets with overlapping but non-identical columns?	[A2]	CO2
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Part B [2 x 11 = 22 Marks]

2.5.	a. You have two datasets: <code>customer_info.csv</code> containing customer demographics and <code>purchase_history.csv</code> containing purchase details. Explain how you would integrate these datasets to analyze the correlation between customer demographics and purchasing behavior. Provide a detailed explanation and example code (e.g., in Python) to demonstrate the integration process. Data Table Example: Customer Info: <table><tr><th>Customer_ID</th><th>Name</th><th>Age</th><th>Gender</th><th>Location</th></tr><tr><td>301</td><td>Alice</td><td>30</td><td>Female</td><td>New York</td></tr><tr><td>302</td><td>Bob</td><td>40</td><td>Male</td><td>Chicago</td></tr><tr><td>303</td><td>Charlie</td><td>25</td><td>Male</td><td>San Francisco</td></tr></table> Data Table Example: Purchase History: <table><tr><th>Purchase ID</th><th>Customer ID</th><th>Purchase Date</th><th>Product ID</th><th>Amount</th></tr><tr><td>P001</td><td>301</td><td>2023-09-01</td><td>PR01</td><td>100.00</td></tr><tr><td>P002</td><td>302</td><td>2023-09-02</td><td>PR02</td><td>150.00</td></tr><tr><td>P003</td><td>303</td><td>2023-09-03</td><td>PR01</td><td>200.00</td></tr></table>	Customer_ID	Name	Age	Gender	Location	301	Alice	30	Female	New York	302	Bob	40	Male	Chicago	303	Charlie	25	Male	San Francisco	Purchase ID	Customer ID	Purchase Date	Product ID	Amount	P001	301	2023-09-01	PR01	100.00	P002	302	2023-09-02	PR02	150.00	P003	303	2023-09-03	PR01	200.00	[B2]	CO2
Customer_ID	Name	Age	Gender	Location																																							
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302	Bob	40	Male	Chicago																																							
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P001	301	2023-09-01	PR01	100.00																																							
P002	302	2023-09-02	PR02	150.00																																							
P003	303	2023-09-03	PR01	200.00																																							
	[OR]																																										
	b. You have a dataset <code>sales_data.csv</code> with records of sales transactions including columns like <code>transaction_id</code>, <code>date</code>, <code>product_id</code>, <code>quantity</code>, and <code>price</code>. The dataset contains errors such as negative quantities, missing prices, and non-unique transaction IDs. Describe the data cleaning steps you would take to prepare this dataset for analysis. Provide a code snippet (e.g., in Python) to illustrate your approach. Data Table Example: Sales Data <table><tr><th>Transaction_ID</th><th>Date</th><th>Product_ID</th><th>Quantity</th><th>Price</th></tr><tr><td>S001</td><td>2023-09-01</td><td>PR01</td><td>5</td><td>50.00</td></tr><tr><td>S002</td><td>2023-09-02</td><td>PR02</td><td>-2</td><td>75.00</td></tr><tr><td>S003</td><td>2023-09-03</td><td>PR01</td><td>3</td><td></td></tr><tr><td>S001</td><td>2023-09-01</td><td>PR01</td><td>5</td><td>50.00</td></tr></table>	Transaction_ID	Date	Product_ID	Quantity	Price	S001	2023-09-01	PR01	5	50.00	S002	2023-09-02	PR02	-2	75.00	S003	2023-09-03	PR01	3		S001	2023-09-01	PR01	5	50.00	[B2]	CO2															
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2.6.	a. Given the dataset kollywood_movies.csv that contains various attributes related to Kollywood movies, perform data discretization on the numerical attributes. Specifically: - Analyze and justify the selection of numerical attributes for discretization. Discuss the chosen binning strategies (e.g., equal-width, equal-frequency, or clustering) for each attribute and the rationale behind these choices. - Implement the discretization process and compare the original continuous data with the discretized data using appropriate visualizations. - Discuss the impact of discretization on data analysis and any patterns or insights that emerge from the discretized dataset. kollywood_movies.csv <table><tr><th>Movie_ID</th><th>Title</th><th>Director</th><th>Genre</th><th>Release_Year</th><th>Budget (INR)</th><th>Box_Office (INR)</th><th>Rating</th><th>Duration (min)</th><th>Actor1</th><th>Actor2</th><th>Language</th><th>Country</th></tr><tr><td>1</td><td>Movie A</td><td>Director A</td><td>Action</td><td>2015</td><td>50000000</td><td>120000000</td><td>7.5</td><td>150</td><td>Actor A</td><td>Actor B</td><td>Tamil</td><td>India</td></tr><tr><td>2</td><td>Movie B</td><td>Director B</td><td>Drama</td><td>2016</td><td>30000000</td><td>80000000</td><td>6.8</td><td>140</td><td>Actor C</td><td>Actor D</td><td>Tamil</td><td>India</td></tr><tr><td>3</td><td>Movie C</td><td>Director C</td><td>Comedy</td><td>2017</td><td>40000000</td><td>90000000</td><td>7.2</td><td>130</td><td>Actor E</td><td>Actor F</td><td>Tamil</td><td>India</td></tr><tr><td>4</td><td>Movie D</td><td>Director D</td><td>Action</td><td>2018</td><td>70000000</td><td>150000000</td><td>8.0</td><td>160</td><td>Actor G</td><td>Actor H</td><td>Tamil</td><td>India</td></tr><tr><td>5</td><td>Movie E</td><td>Director E</td><td>Thriller</td><td>2019</td><td>20000000</td><td>50000000</td><td>6.5</td><td>120</td><td>Actor I</td><td>Actor J</td><td>Tamil</td><td>India</td></tr><tr><td>6</td><td>Movie F</td><td>Director F</td><td>Horror</td><td>2020</td><td>35000000</td><td>85000000</td><td>7.0</td><td>140</td><td>Actor K</td><td>Actor L</td><td>Tamil</td><td>India</td></tr><tr><td>7</td><td>Movie G</td><td>Director G</td><td>Drama</td><td>2021</td><td>60000000</td><td>130000000</td><td>7.8</td><td>150</td><td>Actor M</td><td>Actor N</td><td>Tamil</td><td>India</td></tr><tr><td>8</td><td>Movie H</td><td>Director H</td><td>Action</td><td>2022</td><td>80000000</td><td>200000000</td><td>8.5</td><td>170</td><td>Actor O</td><td>Actor P</td><td>Tamil</td><td>India</td></tr><tr><td>9</td><td>Movie I</td><td>Director I</td><td>Comedy</td><td>2023</td><td>25000000</td><td>70000000</td><td>6.9</td><td>120</td><td>Actor Q</td><td>Actor R</td><td>Tamil</td><td>India</td></tr><tr><td>10</td><td>Movie J</td><td>Director J</td><td>Thriller</td><td>2024</td><td>45000000</td><td>110000000</td><td>7.3</td><td>140</td><td></td><td></td><td></td><td></td></tr></table>	Movie_ID	Title	Director	Genre	Release_Year	Budget (INR)	Box_Office (INR)	Rating	Duration (min)	Actor1	Actor2	Language	Country	1	Movie A	Director A	Action	2015	50000000	120000000	7.5	150	Actor A	Actor B	Tamil	India	2	Movie B	Director B	Drama	2016	30000000	80000000	6.8	140	Actor C	Actor D	Tamil	India	3	Movie C	Director C	Comedy	2017	40000000	90000000	7.2	130	Actor E	Actor F	Tamil	India	4	Movie D	Director D	Action	2018	70000000	150000000	8.0	160	Actor G	Actor H	Tamil	India	5	Movie E	Director E	Thriller	2019	20000000	50000000	6.5	120	Actor I	Actor J	Tamil	India	6	Movie F	Director F	Horror	2020	35000000	85000000	7.0	140	Actor K	Actor L	Tamil	India	7	Movie G	Director G	Drama	2021	60000000	130000000	7.8	150	Actor M	Actor N	Tamil	India	8	Movie H	Director H	Action	2022	80000000	200000000	8.5	170	Actor O	Actor P	Tamil	India	9	Movie I	Director I	Comedy	2023	25000000	70000000	6.9	120	Actor Q	Actor R	Tamil	India	10	Movie J	Director J	Thriller	2024	45000000	110000000	7.3	140					[B2]	CO2
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	b. The following is a dataset that involves child mortality rate and is inspired by data collected from UNICEF. According to the report, the world has achieved substantial success in reducing child mortality during the last few decades. According to the UNICEF report, globally the under-five age mortality rate has decreased from 93 deaths per 1000 live births in 1990 to less than 50 in 2016. However, as you can see, the data set has a number of missing instances, which need to be fixed before a clear progress on child mortality can be explained from the year of 1990 to 2016. Use this data set to complete the following tasks:	[B2]	CO2																																																																																																																																															

	a. Address all the missing values using the techniques at your disposal. b. Prepare the data set to establish the following relations: i. Under-five mortality rate and neonatal mortality rate. ii. Infant mortality rate and neonatal mortality rate. iii. Year and infant mortality rate. [Hints: You may think of converting the mortality rates into five-point Likert scale values. You may count the year before this dataset (i.e., 1989) as the starting point of this program, to assess the progress we have made as the years have passed.]																																																																																																																		
	<table border="1"> <thead> <tr> <th>Year</th><th>Under 5 mortality rate</th><th>Infant mortality rate</th><th>Neonatal mortality rate</th></tr> </thead> <tbody> <tr><td>1990</td><td>93.4</td><td>64.8</td><td>36.8</td></tr> <tr><td>1991</td><td>92.1</td><td>63.9</td><td>36.3</td></tr> <tr><td>1992</td><td>90.9</td><td>63.1</td><td>35.9</td></tr> <tr><td>1993</td><td>89.7</td><td>62.3</td><td>35.4</td></tr> <tr><td>1994</td><td>88.7</td><td>61.4</td><td></td></tr> <tr><td>1995</td><td>87.3</td><td>60.5</td><td>34.4</td></tr> <tr><td>1996</td><td>85.6</td><td>59.4</td><td>33.7</td></tr> <tr><td>1997</td><td></td><td>58.2</td><td>33.1</td></tr> <tr><td>1998</td><td>82.1</td><td>56.9</td><td>32.3</td></tr> <tr><td>1999</td><td>79.9</td><td>55.4</td><td>31.5</td></tr> <tr><td>2000</td><td>77.5</td><td>53.9</td><td>30.7</td></tr> <tr><td>2001</td><td>74.8</td><td>52.1</td><td>29.8</td></tr> <tr><td>2002</td><td>72</td><td></td><td>28.9</td></tr> <tr><td>2003</td><td>69.2</td><td>48.6</td><td>28</td></tr> <tr><td>2004</td><td>66.7</td><td>46.9</td><td></td></tr> <tr><td>2005</td><td></td><td>45.1</td><td>26.1</td></tr> <tr><td>2006</td><td>61.1</td><td>43.4</td><td>25.3</td></tr> <tr><td>2007</td><td>58.5</td><td></td><td>24.4</td></tr> <tr><td>2008</td><td>56.2</td><td>40.3</td><td>23.6</td></tr> <tr><td>2009</td><td>53.7</td><td>38.8</td><td>22.9</td></tr> <tr><td>2010</td><td></td><td>37.4</td><td>22.2</td></tr> <tr><td>2011</td><td>49.3</td><td>36</td><td>21.5</td></tr> <tr><td>2012</td><td>47.3</td><td>34.7</td><td>20.8</td></tr> <tr><td>2013</td><td>45.5</td><td>33.6</td><td>20.2</td></tr> <tr><td>2014</td><td>43.7</td><td></td><td>19.6</td></tr> <tr><td>2015</td><td>42.2</td><td>31.4</td><td>19.1</td></tr> <tr><td>2016</td><td>40.8</td><td>30.5</td><td>18.6</td></tr> </tbody> </table>	Year	Under 5 mortality rate	Infant mortality rate	Neonatal mortality rate	1990	93.4	64.8	36.8	1991	92.1	63.9	36.3	1992	90.9	63.1	35.9	1993	89.7	62.3	35.4	1994	88.7	61.4		1995	87.3	60.5	34.4	1996	85.6	59.4	33.7	1997		58.2	33.1	1998	82.1	56.9	32.3	1999	79.9	55.4	31.5	2000	77.5	53.9	30.7	2001	74.8	52.1	29.8	2002	72		28.9	2003	69.2	48.6	28	2004	66.7	46.9		2005		45.1	26.1	2006	61.1	43.4	25.3	2007	58.5		24.4	2008	56.2	40.3	23.6	2009	53.7	38.8	22.9	2010		37.4	22.2	2011	49.3	36	21.5	2012	47.3	34.7	20.8	2013	45.5	33.6	20.2	2014	43.7		19.6	2015	42.2	31.4	19.1	2016	40.8	30.5	18.6		
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Answer ALL Questions
Part A [2 x 2 = 4 Marks]

3.1.	Evaluate the role of visual aids such as histograms, box plots, and scatter plots in identifying underlying patterns in the data. Which visual aid would be most effective in spotting outliers and why?	[A2]	CO3
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3.2.	Compare the effectiveness of EDA techniques in identifying data quality issues versus simply relying on automated data validation tools. What are the advantages of using EDA?	[A2]	CO3
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Part B [1 x 11 = 11 Marks]

3.3.	a.	Evaluate the role of visual aids such as histograms, box plots, and scatter plots in identifying underlying patterns in the data. Which visual aid would be most effective in spotting outliers and why?	[B2]	CO3																																																																																																																
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	b.	You are provided with a dataset <code>students_performance.csv</code> that contains information about students' demographic data and their scores in three subjects: Math, Reading, and Writing. Perform an exploratory data analysis to uncover patterns and insights in the data. Specifically: - Identify key demographic factors that influence students' performance in each subject. - Spot any trends or anomalies in the students' scores. - Compare the effectiveness of using EDA versus relying solely on summary statistics for identifying these insights. - Provide visualizations and a detailed explanation of your findings. <code>students_performance.csv</code> <table><tr><th>Student_ID</th><th>Age</th><th>Gender</th><th>Parental_Level_of_Education</th><th>Math_Score</th><th>Reading_Score</th><th>Writing_Score</th></tr><tr><td>1</td><td>15</td><td>Female</td><td>Bachelor's degree</td><td>65</td><td>70</td><td>72</td></tr><tr><td>2</td><td>16</td><td>Male</td><td>Some college</td><td>78</td><td>82</td><td>80</td></tr><tr><td>3</td><td>15</td><td>Female</td><td>High school</td><td>90</td><td>88</td><td>93</td></tr><tr><td>4</td><td>16</td><td>Male</td><td>Associate's degree</td><td>55</td><td>58</td><td>60</td></tr><tr><td>5</td><td>15</td><td>Female</td><td>Bachelor's degree</td><td>72</td><td>75</td><td>78</td></tr><tr><td>6</td><td>16</td><td>Male</td><td>Some college</td><td>85</td><td>87</td><td>89</td></tr><tr><td>7</td><td>15</td><td>Female</td><td>High school</td><td>92</td><td>95</td><td>97</td></tr><tr><td>8</td><td>16</td><td>Male</td><td>Associate's degree</td><td>60</td><td>62</td><td>64</td></tr><tr><td>9</td><td>15</td><td>Female</td><td>Bachelor's degree</td><td>68</td><td>70</td><td>72</td></tr><tr><td>10</td><td>16</td><td>Male</td><td>Some college</td><td>80</td><td>82</td><td>85</td></tr><tr><td>11</td><td>15</td><td>Female</td><td>High school</td><td>93</td><td>95</td><td>97</td></tr><tr><td>12</td><td>16</td><td>Male</td><td>Associate's degree</td><td>58</td><td>60</td><td>62</td></tr><tr><td>13</td><td>15</td><td>Female</td><td>Bachelor's degree</td><td>75</td><td>78</td><td>80</td></tr><tr><td>14</td><td>16</td><td>Male</td><td>Some college</td><td>82</td><td>85</td><td>87</td></tr><tr><td>15</td><td>15</td><td>Female</td><td>High school</td><td>95</td><td>98</td><td>100</td></tr></table>	Student_ID	Age	Gender	Parental_Level_of_Education	Math_Score	Reading_Score	Writing_Score	1	15	Female	Bachelor's degree	65	70	72	2	16	Male	Some college	78	82	80	3	15	Female	High school	90	88	93	4	16	Male	Associate's degree	55	58	60	5	15	Female	Bachelor's degree	72	75	78	6	16	Male	Some college	85	87	89	7	15	Female	High school	92	95	97	8	16	Male	Associate's degree	60	62	64	9	15	Female	Bachelor's degree	68	70	72	10	16	Male	Some college	80	82	85	11	15	Female	High school	93	95	97	12	16	Male	Associate's degree	58	60	62	13	15	Female	Bachelor's degree	75	78	80	14	16	Male	Some college	82	85	87	15	15	Female	High school	95	98	100	[B2]	CO3
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